Minimal-Shot AV Evidence Deck

SOTA Submission Results + CoRL Transfer Diagnostics

Alba Maria Tellez Fernandez

Waymo Open Dataset E2E Challenge / SOTA + CoRL 2027 branch

Two Audiences, One Evidence Package

This deck supports two related but distinct uses.

Audience	Emphasis	Boundary
SOTA audit	working system, bundles, WOD validation-CV	not production
CoRL paper	transfer failure diagnosis	not method dominance

Do not collapse them into one claim.

The SOTA submission is a system/evidence package. The CoRL story is a mechanistic diagnosis built on that infrastructure.

SOTA Submission Scope

Primary SOTA track:

- episode-free Spotlight Reflex closed-loop runtime
- randomized WOD-style long-tail scenarios
- explicit minimal-shot integrity audit
- no public-road deployment or legal certification claim

Auxiliary benchmark track:

- WOD-E2E candidate-selection harness
- segment-grouped validation-CV only
- frozen Cosmos / InternVLA grounding ablations

Source: `artifacts/sota_judging_criteria_audit.json`,

SOTA Readiness Gates

Gate	Status
Grand archive ready	pass
Minor archive ready	pass
Minimal-shot integrity	pass
Judging-criteria evidence	pass
Production no-go boundary declared	pass
WOD symbolic-trust risk-control boundary	pass

No blockers are reported in `artifacts/final_submission_readiness_audit.json`.

The submission boundary is explicit: `submission_candidate_not_hidden_test_not_production`.

SOTA Primary Simulation Result

Closed-loop simulation headline from the existing SOTA evidence package:

Suite	Runs	Collisions	Pass
WOD-style, all 11 clusters	110	0	100%
Compositional OOD	60	0	100%
Adversarial	60	0	100%
Hidden holdout	60	0	100%
Gauntlet	60	0	60%
Total	350	0	93.1%

COMPASS: **9.137 / 10** over 700 ranked runs.

Source: `README.md`, `artifacts/compass_evidence_report.json`

SOTA Baseline Comparison

Matched gauntlet comparison: same 420 scenarios, same seeds.

Policy	Pass rate	Collision rate
Baseline, no world-state reasoning	2.1%	20.5%
Spotlight Reflex	**57.6%**	**7.9%**

Interpretation for SOTA:

- the system contribution is not just a score
- it is an auditable geometry-grounded runtime path
- failures are analyzed rather than hidden behind an aggregate metric

Source: `README.md`, `docs/presentation.tex`.

WOD-E2E Auxiliary Benchmark

479 WOD-E2E validation frames, segment-grouped CV.

Selector	RFS	Folds	Role
Waymo baseline	7.022	-	reference
Local baseline	7.131	-	reference
Gate-only	7.803	5	selector baseline
RFF direct policy	7.834	5	learned baseline
GPU MLP + Cosmos 64d	**7.845**	5	champion validation-CV
Oracle selector	9.264	5	upper bound

This is auxiliary evidence, not a hidden-test or zero-shot WOD claim.

Source: `README.md`, `artifacts/wod_grounding_ablation_table.md`

CoRL Research Claim

Internal closed-loop success is not a reliable transfer proxy.

We find that simulator transfer failure decomposes into separate axes:

- geometry: offroad and route reconstruction
- safety: collision and proximity violation
- progress: distance and route completion
- lane adherence: wrong-lane or lane-violation behavior

No evaluated intervention is co-monotone across all axes in AlpaSim.

Why This Matters Now

Recent VLA work moves toward physically grounded latent reasoning.

LaST-VLA argues that autonomous-driving VLA reasoning should use 3D geometric priors
and world-model dynamics, not only text chain-of-thought.

It reports NAVSIM v1/v2 gains through latent spatio-temporal reasoning.

Our question is narrower and diagnostic:

If a policy or selector is internally strong, which transfer axes actually survive when the observation and dynamics interface changes?

Reference: LaST-VLA, arXiv:2603.01928

Evidence Stack

Layer	Purpose	Evidence
SOTA runtime	Working minimal-shot AV system	Spotlight Reflex, COMPASS, submission bundles
Custom simulator	Controlled long-tail AV stress lab	LHS generator, DAgger, proxy perturbations
AlpaSim adapter	External sim-to-sim validation on WOD-E2E clips	10 matched clips, partial 13-scene refresh
WOD candidate-selection	Real-data grounding probe	479 validation frames, segment-grouped CV

The tracks are deliberately separated.

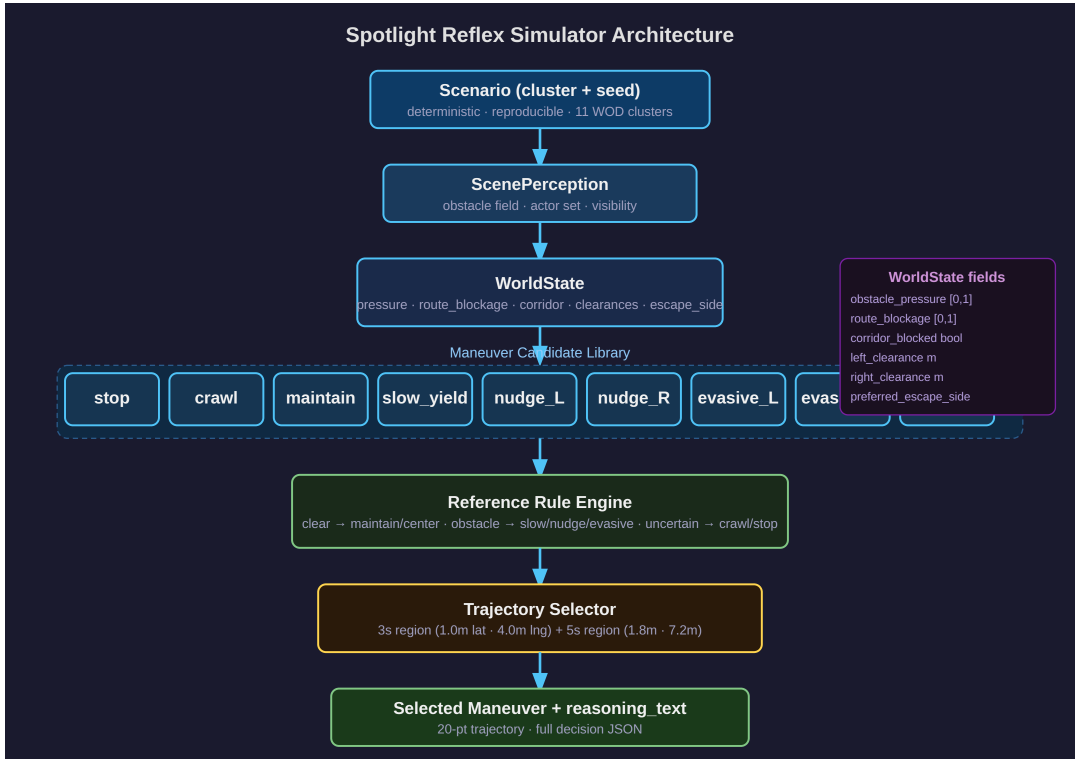
Simulator metrics are not used to tune WOD selector results, and WOD validation

What the Simulator Tests

The internal simulator is a controlled micro-lab, not a realism claim.

It lets us isolate:

- discrete token action spaces
- rule-based geometric selection
- BC and DAgger imitation variants
- proxy-state corruption with true-state evaluation
- matched seeds across policies and perturbations



Visual Failure and Success Cases

Baseline failure	Geometry-grounded success
	

Same scenario family: wrong-way actor under low visibility.

The baseline extrapolates into the hazard. Spotlight selects an evasive token from

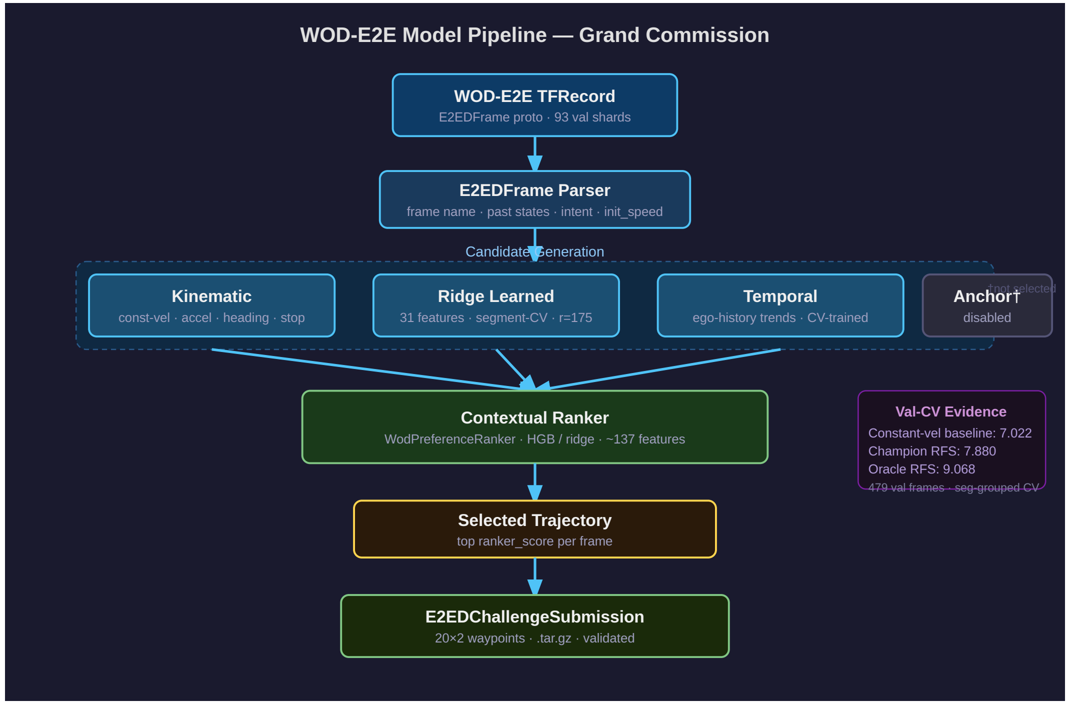
Policy Interface

State:

- 10 scalar geometric features for learned policies
- six core world-state scalars for Spotlight

Action:

- 9 ManeuverTokens
- stop, crawl, maintain, slow-yield
- nudge left/right, evasive left/right, lane-recover



The policy selects a token; the

Internal Result: DAgger Closes the Local Frontier

Held-out Latin-hypercube sweep: 12 profiles, seeds 1-10, Gauntlet / Adversarial / Hidden.

Each agent: 1080 closed-loop rollouts.

Agent	Overall pass / PV	Gauntlet	Adversarial	Hidden
Continuous-BC	77.69 / 3.52	77.50 / 4.31	75.42 / 2.08	83.33 / 1.67
Token-BC	76.85 / 4.26	77.64 / 4.86	71.67 / 2.92	82.50 / 3.33
		79.17 /		81.67 /

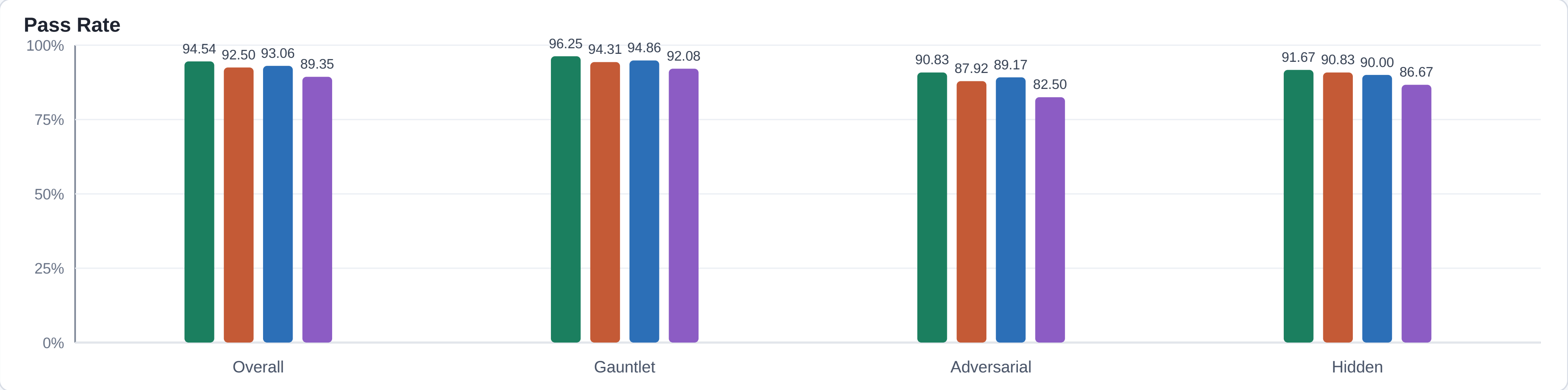
Internal Result: More DAgger Is Not Monotonic

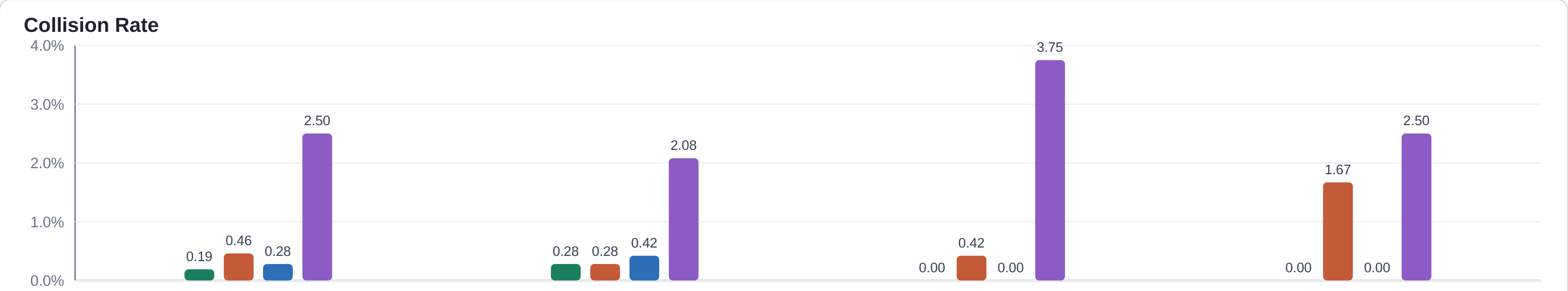
Policy	Overall pass / PV	Gauntlet	Adversarial	Hidden
Token-DAgger, 1-step	93.98 / 0.46	96.81 / 0.56	87.50 / 0.42	90.00 / 0.00
Token-DAgger, 2-step	**94.54 / 0.19**	**96.25 / 0.28**	**90.83 / 0.00**	**91.67 / 0.00**
Token-DAgger, 3-step	92.50 / 0.46	94.31 / 0.28	87.92 / 0.42	90.83 / 1.67
3-step source decay	93.06 / 0.28	94.86 / 0.42	89.17 / 0.00	90.00 / 0.00
3-step no inverse	89.35 / 2.50	92.08 /	82.50 / 3.75	86.67 /

DAgger Aggregation Visual

DAgger Aggregation Ablations on Held-Out Latin-Hypercube Sweep

Two-step DAgger remains the frontier. Source decay partially recovers the three-step collapse; removing inverse-frequency weights breaks closed-loop safety.





Negative Result: Scoring Ego Trajectories Was Not Enough

We trained trajectory-informed scorers over candidate futures with:

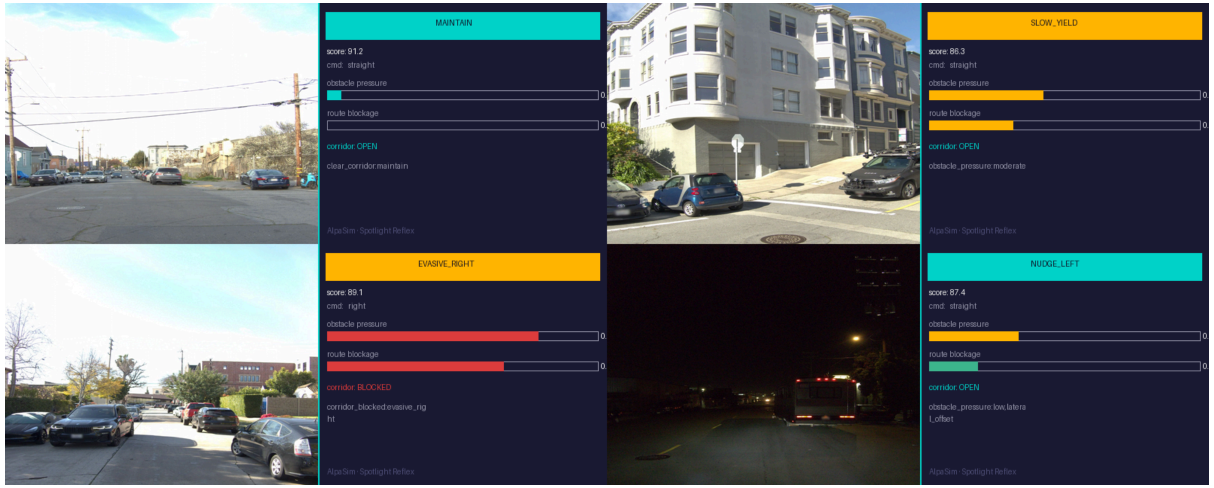
- curvature and heading deltas
- path length, acceleration, jerk proxies
- minimum dynamic clearance and time-to-closest-approach
- signed lateral and longitudinal miss distances

Policy	Gauntlet pass / PV	Adversarial	Hidden
Interaction scorer, learner states	86.81 / 1.94	85.00 / 1.67	89.17 / 1.67

External Diagnostic: AlpaSim

AlpaSim uses real WOD-E2E front-camera frames and an adapter that reconstructs:

- brightness
- route command
- ego dynamics
- sparse hazard primitives
- geometric scalars consumed by the policy



This is a transfer diagnostic, not a

AlpaSim Bridge

The external bridge keeps the policy binary fixed.

What changes is the interface:

- camera and route observations become proxy geometric scalars
- sparse hazards become obstacle and actor primitives
- token selection is evaluated under AlpaSim execution

This is exactly where sim-to-sim mismatch is exposed.

AlpaSim Matrix: One Fix Does Not Repair All Axes

10 matched WOD-E2E clips completed by all four variants.

Model	Collision	Offroad	Wrong lane	Progress	Dist. m	Dist.-GT
Raw DAgger iter2	0.600	0.900	0.700	0.034	61.98	13.86
Clamped lateral	0.700	0.500	0.200	0.384	59.85	5.21
Hard veto hybrid	0.800	0.200	0.700	0.837	166.06	22.71
Source decay	0.600	0.900	0.400	0.175	62.92	27.16

What the AlpaSim Interventions Show

Intervention	Repairs	Worsens or fails
Clamped lateral	offroad, wrong-lane, progress, distance-to-GT	collision increases
Hard veto hybrid	progress, offroad	collision increases, wrong-lane unchanged
Source decay	wrong-lane, progress	offroad unchanged, distance-to-GT worsens

The hard veto is especially diagnostic:

- vetoed 1987 / 1990 DAgger argmax decisions
- selected mostly the geometric scorer path
- recovered progress by collapsing toward Spotlight-style selection

Controlled Proxy-State Perturbation

Internal simulator, true-state physics and evaluation fixed.

Only the planner-visible proxy state is corrupted.

Perturbation	Raw p/c/l	Clamp	Hybrid	
clean	0.917/0.083/0.312	0.938/0.062/0.167	0.958/0.042/0.167	0.938/
heading bias	0.896/0.104/0.562	0.938/0.062/0.396	0.938/0.062/0.375	0.896/
actor latency	0.604/0.396/0.875	0.604/0.396/0.521	**0.854/0.146/0.208**	0.833/
route offset	0.938/0.062/0.229	0.875/0.125/0.021	0.896/0.104/0.062	0.938/

Source: `artifacts/internal_proxy_transfer_medium.md`.

Why the Proxy Sweep Matters

AlpaSim alone leaves an adapter-confound objection.

The proxy sweep removes that escape hatch:

- same simulator dynamics
- same token library
- same controller
- matched scenarios
- only proxy state is corrupted

The same multi-axis tradeoff appears internally under controlled corruption.

That makes the external result mechanistic rather than anecdotal.

The Proposition

Let each candidate token have a metric vector:

`m(a) = [collision risk, offroad risk, lane risk, progress loss, tracking error]`

If a proxy-state perturbation changes the action ordering for one metric but not another,
a scalar intervention can improve one axis while worsening another.

The empirical question is not whether this is possible.

The empirical question is whether it occurs repeatedly under realistic AV transfer interfaces.

Our answer: yes, in AlpaSim and in controlled proxy perturbations.

WOD Grounding Probe

479 WOD-E2E validation frames, segment-grouped CV.

Ablation	RFS	Oracle	Regret	Oracle match	Grounding signal
Scalar / geometry only	7.695	9.068	1.373	0.403	none
InternVLA only	7.672	9.225	1.554	0.392	InternVLA
Cosmos + InternVLA linear fusion	7.728	9.208	1.480	0.403	Cosmos + InternVLA
Cosmos 64d nonlinear head	**7.845**	9.264	1.419	0.390	Cosmos + nonlinear head

Grounding helps only with the right head

CoRL Evidence Strength Audit

Current CoRL audit conclusion: `strong_diagnostic`.

Gate	Current	Target	Status
Internal proxy diagnostic	288 cases	>=288 cases	pass
External transfer diagnostic	10 matched scenes	>=10 scenes	pass
Larger external scale	10 scenes	>=30 scenes	not yet
Uniform positive method	0 external dominance cases	>=1	not yet
Grounding prior signal	+0.150 RES	>0	pass

SOTA vs CoRL Claim Boundary

Question	Answer
Is the SOTA submission packaged?	yes, readiness audit has no blockers
Is this a production AV safety claim?	no, production audit is `no_go`
Is WOD 7.845 a hidden-test claim?	no, validation-CV only
Is the CoRL method a uniform winner?	no, external dominance count is 0
Is the CoRL diagnosis supported?	yes, internal proxy + AlpaSim both show tradeoffs

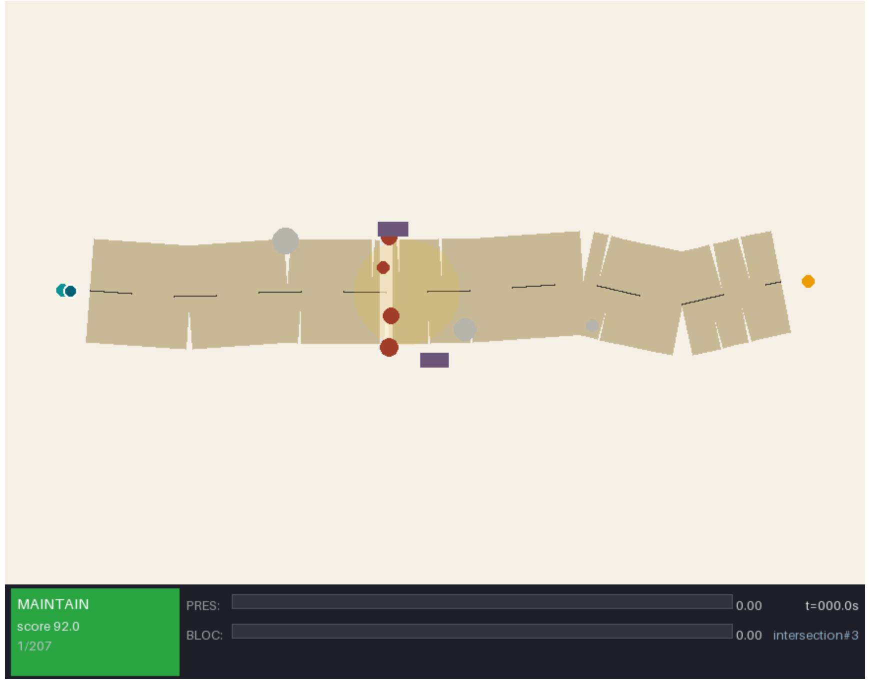
This boundary should stay visible in the presentation.

Media and Figures Included

Asset	Use
`docs/images/baseline_spotlight.gif`	baseline visual failure
`docs/images/spotlight_success.gif`	geometry-grounded success
`docs/images/construction_success.gif`	long-tail construction case
`docs/images/intersection_stress.gif`	multi-actor stress case
`docs/images/alpasim_reasoning_panel.png`	AlpaSim adapter reasoning
`docs/images/dagger_aggregation_ablation.svg`	DAgger aggregation plot
`docs/images/alpasim-bridge.svg`	external bridge diagram

GIFs are used as short videos in GitHub, Marp, and most slide renderers.

Additional Visual Cases

Construction	Intersection stress
	

These examples are not the proof.

They show what the controlled simulator is stressing: narrow corridors, route

Reproducibility Path for Auditors

SOTA submission audits:

```
artifacts/final_submission_readiness_audit.json
artifacts/sota_judging_criteria_audit.json
artifacts/minimal_shot_claim_audit.json
artifacts/sota_submission_bundles/
```

CoRL audit wrapper:

```
./scripts/run_corl2027_audit.sh
```

Core evidence files:

```
docs/corl2027/paper.tex
docs/corl2027/AUDIT.md
artifacts/corl_evidence_strength_audit.md
artifacts/internal_proxy_transfer_medium.md
```

Submission Framing

For SOTA, frame this as:

“A reproducible minimal-shot AV system with explicit claim boundaries, simulation evidence,
WOD validation-CV auxiliary results, and packaged audit artifacts.”

For CoRL, do not frame this as:

“Our hybrid method dominates.”

The data does not support that.

Frame the CoRL result as:

“Internal imitation-learning success can hide transfer-axis disagreement.

We provide a controlled diagnostic, reproduce the effect externally in AlpaSim,

Takeaway

The surprising result is not that DAgger fails externally.

The result is where it fails:

- the internal simulator says the learned token policy is calibrated
- AlpaSim shows the selected tokens are often externally incompatible
- controlled proxy corruption reproduces the same axis conflicts
- frozen world/VLA priors help only through head-sensitive grounded selection

Robust AV evaluation needs per-axis transfer diagnostics, not only offline accuracy or aggregate closed-loop pass rate.